

# ORGNZ-02: Understanding Grail Buckets

**Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 2 of 10 | **Created:** January 2026 | **Last Updated:** 05/06/2026

## Overview

**Buckets** are logical storage containers in Dynatrace Grail where records are stored. Think of a bucket as a folder in a filesystem. Use buckets to group telemetry data that naturally belongs together, such as data from the same region, environment, or with the same sensitivity classification.

## Prerequisites

| Requirement                  | Details                                                            |
|------------------------------|--------------------------------------------------------------------|
| <b>Dynatrace Environment</b> | SaaS environment with Grail enabled                                |
| <b>Permissions</b>           | <code>storage:buckets:read</code> , <code>storage:logs:read</code> |
| <b>Knowledge</b>             | Completed ORGNZ-01 (Introduction)                                  |
| <b>Data</b>                  | At least 1 hour of log data across default buckets                 |

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## Bucket Administration Permissions

| Permission                                       | Purpose                    |
|--------------------------------------------------|----------------------------|
| <code>storage:bucket-definitions:read</code>     | View bucket configurations |
| <code>storage:bucket-definitions:write</code>    | Create and modify buckets  |
| <code>storage:bucket-definitions:delete</code>   | Remove buckets             |
| <code>storage:bucket-definitions:truncate</code> | Truncate bucket data       |

## Learning Objectives

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By the end of this notebook, you will:

- Understand bucket fundamentals and supported data types
- Know bucket limits and query constraints
- Query bucket information using DQL
- Recognize when custom buckets are needed

## What Are Buckets?

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Buckets organize data to meet performance, compliance, and retention requirements:

| Function             | Description                                      |
|----------------------|--------------------------------------------------|
| <b>Data Grouping</b> | Organize large datasets into meaningful segments |

| Function                        | Description                                                 |
|---------------------------------|-------------------------------------------------------------|
| <b>Streamlined Organization</b> | Records that should be handled together are stored together |
| <b>Query Performance</b>        | Reduced queried dataset improves query speed                |
| <b>Retention Management</b>     | Each bucket has its own retention policy                    |
| <b>Access Control</b>           | Bucket-level IAM policies for team isolation                |
| <b>Cost Attribution</b>         | Enable data cost tracking by organizational unit            |

## Built-in Grail Buckets

Dynatrace provides a set of predefined built-in buckets that cannot be modified. Use this DQL to discover them all:

```
fetch dt.system.buckets
| filter startsWith(name, "default_") or startsWith(name, "dt_")
```

| Bucket Name                    | Table               | Default Retention |
|--------------------------------|---------------------|-------------------|
| default_logs                   | logs                | 35 days           |
| default_metrics                | metrics             | 15 months         |
| default_spans                  | spans               | 10 days           |
| default_events                 | events              | 35 days           |
| default_bizevents              | bizevents           | 35 days           |
| default_securityevents         | security.events     | 1 year            |
| default_securityevents_builtin | security.events     | 3 years           |
| default_user_events            | user.events         | 35 days           |
| default_user_sessions          | user.sessions       | 35 days           |
| default_mobile_user_replays    | mobile.user.replays | 35 days           |
| default_web_user_replays       | web.user.replays    | 35 days           |

| Bucket Name                       | Table                     | Default Retention |
|-----------------------------------|---------------------------|-------------------|
| default_synthetic_events          | synthetic.events          | 35 days           |
| default_synthetic_user_events     | synthetic.user.events     | 35 days           |
| default_synthetic_user_sessions   | synthetic.user.sessions   | 35 days           |
| default_synthetic_detailed_events | synthetic.detailed.events | 35 days           |
| default_application_snapshots     | application.snapshots     | 10 days           |
| dt_system_events                  | dt.system.events          | 1 year            |

**Note:** The `default_securityevents` table was recently migrated to a new Grail security events schema. If you use security events, follow the [Grail security table migration guide](#) to complete any required actions.

**Extended retention:** You can extend bucket retention beyond the defaults by joining the Dynatrace preview program. See [Preview releases](#) for details.



## Supported Data Types (Tables)

Each bucket is associated with exactly **one data type**. You cannot mix data types in the same bucket.

**Custom buckets can be created for:** logs, events, security events, bizevents, and spans.



**Important:** Metrics do not support custom buckets — all metric data routes to `default_metrics`.

## System Tables

In addition to regular data tables, Grail exposes **system tables** — virtual tables representing Grail metadata, not stored in any bucket:

| System Table                        | Contents                                             |
|-------------------------------------|------------------------------------------------------|
| <code>dt.system.buckets</code>      | Bucket inventory (name, retention, size, table type) |
| <code>dt.system.data_objects</code> | Schema and data object metadata                      |
| <code>dt.system.files</code>        | Lookup files uploaded for DQL enrichment             |
| <code>dt.system.events</code>       | Audit events (bucket changes, platform actions)      |

System tables are queried with `fetch dt.system.*` — they never need a `bucket:` parameter.

# Bucket Limits

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## Environment Limits

| Limit                           | Value                    | Notes                               |
|---------------------------------|--------------------------|-------------------------------------|
| Maximum buckets per environment | 80                       | Default limit; can request increase |
| Typical capacity                | Up to 5 TB/day per table | With default bucket limit           |

## Bucket Size Guidelines

| Metric                    | Value      | Impact                                    |
|---------------------------|------------|-------------------------------------------|
| Optimal ingest per bucket | ~1 TB/day  | Best query performance                    |
| Acceptable ingest         | 1-2 TB/day | Reduced query window                      |
| Split threshold           | >2 TB/day  | Dynatrace recommends splitting above this |

## Retention Limits

| Limit             | Value                           |
|-------------------|---------------------------------|
| Minimum retention | 1 day                           |
| Maximum retention | 3,657 days (~10 years + 1 week) |

# Query Constraints

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Understanding query limits is critical for bucket planning:

## Query Limits

| Limit                | Value  | Impact                       |
|----------------------|--------|------------------------------|
| Maximum data scanned | 500 GB | Limits queryable time window |

| Limit                    | Value | Impact                               |
|--------------------------|-------|--------------------------------------|
| Maximum records returned | 1,000 | Use aggregations for larger datasets |
| Maximum response payload | 1 MB  | Large result sets may be truncated   |

## Queryable Window by Ingest Volume

The 500 GB scan limit determines how far back you can query:

| Daily Ingest | Queryable Window |
|--------------|------------------|
| 500 GB/day   | ~24 hours        |
| 1 TB/day     | ~12 hours        |
| 3 TB/day     | ~4 hours         |
| 10 TB/day    | ~1 hour          |

**Implication:** High-volume buckets may only allow querying recent data. Use aggregations, time filters, and strategic bucket partitioning to work within these limits.

## Default Buckets

Dynatrace provides default buckets with varying retention:

| Default Bucket               | Data Type | Default Retention | Notes                       |
|------------------------------|-----------|-------------------|-----------------------------|
| <code>default_logs</code>    | logs      | 35 days           | Standard log retention      |
| <code>default_metrics</code> | metrics   | 15 months         | Extended for trend analysis |
| <code>default_spans</code>   | spans     | 10 days           | Short-term APM data         |

| Default Bucket                              | Data Type       | Default Retention | Notes                    |
|---------------------------------------------|-----------------|-------------------|--------------------------|
| <code>default_events</code>                 | events          | 35 days           | Platform events          |
| <code>default_bizevents</code>              | bizevents       | 35 days           | Business events          |
| <code>default_securityevents</code>         | security_events | 1 year            | Security events          |
| <code>default_securityevents_builtin</code> | security_events | 3 years           | Built-in security events |

**Note:** Query your environment's `dt.system.buckets` to verify current retention settings.

## Querying Bucket Information

## Querying Data from Buckets

## Bucket Characteristics

### Immutability Rules

| Characteristic   | Mutable? | Notes                                 |
|------------------|----------|---------------------------------------|
| Bucket name      | No       | Cannot be changed after creation      |
| Display name     | Yes      | Can be updated anytime                |
| Retention period | Yes      | <b>Caution:</b> Lowering deletes data |
| Data table type  | No       | Fixed at creation                     |



## Data Handling

| Operation                 | Supported? | Alternative                     |
|---------------------------|------------|---------------------------------|
| Move data between buckets | No         | Route new data via OpenPipeline |
| Selective record deletion | No         | Wait for retention to expire    |
| Bucket deletion           | Yes        | Deletes ALL data permanently    |

## Bucket Naming Rules

| Rule               | Requirement                                  |
|--------------------|----------------------------------------------|
| Name length        | 3–100 characters                             |
| First character    | Must be a letter                             |
| Allowed characters | Lowercase alphanumeric, underscores, hyphens |
| Case               | Lowercase only                               |
| Reserved names     | Cannot use <code>default_*</code> prefix     |

## Valid Examples

```
team_logs_30d
finance-audit-logs
prod_metrics_90d
app123_spans
```

## Invalid Examples

```
123_logs      # Starts with number
Team_Logs     # Contains uppercase
default_custom # Uses reserved prefix
my logs       # Contains space
```

# When to Create Custom Buckets

| Need                         | Solution                               |
|------------------------------|----------------------------------------|
| Different retention periods  | Create bucket with specific retention  |
| Cost attribution by team/LOB | Separate buckets per cost center       |
| Simple team isolation        | Bucket-level IAM policies              |
| Regulatory compliance        | Dedicated buckets for compliance data  |
| Query performance            | Isolate high-volume data sources       |
| Debug/verbose logs           | Short-retention bucket to reduce costs |

## Next Steps

Continue with the ORGNZ series:  
- **ORGNZ-03**: Bucket Strategy and Design

## References

- [Grail Buckets](#)
- [Data Retention](#)
- [DQL Reference](#)

*This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*

## DQL: Built-in Bucket Discovery

Discover all built-in Grail buckets in your environment:

```
// List all built-in Grail buckets – inventory check
fetch dt.system.buckets
| filter startsWith(name, "default_") or startsWith(name, "dt_")
| fields name, display_name, dt.system.table, retention_days
| sort dt.system.table asc, name asc
```

## DQL: Bucket Audit Events

Track who created, updated, truncated, or deleted a bucket:

```
// Audit bucket management actions – returns create, update, truncate, delete
events
fetch dt.system.events
| filter event.kind == "AUDIT_EVENT" and event.category ==
"BUCKET_MANAGEMENT"
| sort timestamp desc
```

## DQL: Querying Data from Buckets

Use the `bucket:` parameter to target specific storage:

```
// Query logs from a specific bucket – use the bucket: parameter to target
storage
fetch logs, from:-1h, bucket:"default_logs"
| limit 10
```

```
// Query from multiple buckets simultaneously – controls scan scope
fetch logs, from:-1h, bucket:{"default_logs", "audit_logs", "security_logs"}
| limit 100
```